Unit 07: Confidence Interval

Instructor: Dr. Md Mutasim Billah

Example 1

Let us consider the following situation. Suppose the sample mean $\bar{X}$ is normally distributed, either because the population is normal or because the sample size is large ($n \geq 30$). Then

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right).$$

Assume that the population standard deviation σ is known.

(a) Construct a 95% confidence interval for μ.

(b) Using the result from part (a), can we generalize our answer to construct a $(1-\alpha)100\%$ confidence interval for μ when σ is known?

Example 2: Confidence Interval when σ is known

We are interested in μ, the population mean weight among all CWRU students. Suppose $n = 30$, $\bar{x} = 150$ pounds, and $\sigma = 25$ (known). Construct a 95% confidence interval for μ. *Note:* You do NOT need to calculate exact values.

Example 3: True/False Game: T-P-S

Now that we have constructed a 95% confidence interval for the population mean weight among all CWRU students, what does it actually mean?
Make sure you synchronize with your group members and work through the following interpretations. Discuss whether each statement is **True** or **False**, and be ready to share your reasoning during our "Think–Pair–Share activity"!

(a) We sampled 95% of all CWRU students.

(b) We know that $\mu = 150$.

(c) 95% of CWRU students weight somewhere between 141.05 & 158.95.

(d) We know that $141.05 < \mu < 158.95$.

(e) Based on our sample, values of μ between 141.05 and 158.95 are plausible for the population mean weight of CWRU students at the 95% confidence level.

(f) $P(141.05 < \mu < 158.95) = 0.95$.

(g) We are 95% confident that $141.05 < \mu < 158.95$.

(h) If this sampling procedure were repeated many times and an interval were constructed using the same method each time, approximately 95% of those intervals would contain the true population mean.

Example 4: Analogy: Horseshoes and Confidence Intervals

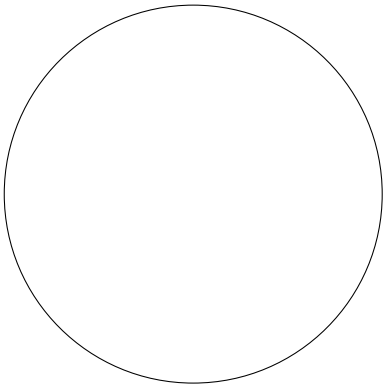


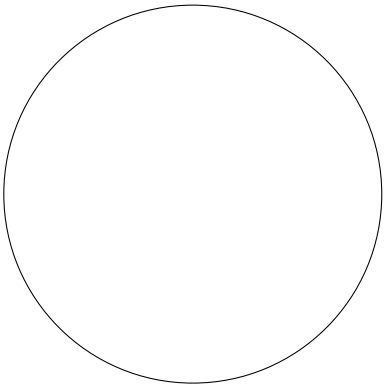
In the horseshoe analogy, the thrower cannot see or hear the pin, just as a sample has no direct knowledge of the true population mean μ.
Each horseshoe represents a confidence interval, and the pin represents the true parameter μ.

Example 5: Analogy: Coin Toss and Confidence Intervals



Imagine standing near Wade Lagoon by the Cleveland Museum of Art and tossing a special coin from a bridge toward the water!





Example 6: Understanding Confidence Interval

(a) Consider the confidence interval formula for μ when σ is known:

$$\bar{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

Without performing calculations, discuss what would make the interval **wider** or **thinner**.

(b) Would a 100% confidence interval for μ make sense? If we tried to construct a 100% confidence interval, what would happen to the critical value and the width of the interval? Explain why such an interval may or may not be useful in practice.

Example 7: Margin of Error & Sample Size

Recall the margin of error (ME) is

$$\text{ME} \;=\; z_{\alpha/2}\,\frac{\sigma}{\sqrt{n}}.$$

(a) If we want the margin of error to equal some small number B, what sample size is necessary?

$$z_{\alpha/2}\,\frac{\sigma}{\sqrt{n}} = B.$$

Solving for n:

$$\frac{\sigma}{\sqrt{n}} = \frac{B}{z_{\alpha/2}} \quad\Longrightarrow\quad \sqrt{n} = \frac{z_{\alpha/2}\sigma}{B} \quad\Longrightarrow\quad n = \left(\frac{z_{\alpha/2}\sigma}{B}\right)^2.$$

(b) Following **example 2**, 95% confidence interval for μ was 150 ± 8.95, when $\sigma = 25$ (known). If we want the margin of error to equal 3 pounds with 95% confidence, what should be the required sample size of CWRU students?

Example 8: Other Confidence Levels

Using the same setup as in **Example 1**, how would we construct confidence intervals for μ (when σ is known) with different confidence levels, such as 90% and 99%?

Example 9: Confidence Interval of μ when σ is unknown

Suppose μ = population mean cholesterol level among 70-year-old American men. Let $n = 19$ (small n, so we must assume the population is normal), $\bar{x} = 44.2$, σ is unknown, and $s = 3.8$.

(a) Find a 90% confidence interval for μ. *Note:* You do NOT need to calculate exact values.

(b) Write down an appropriate interpretation of your finding in part (a) in context.

Example 10: Reverse-engineer

Suppose $n = 16$, σ is unknown, and an 80% confidence interval for μ was $(48, 54)$. What were $\bar{x}$ and s?

Example 11: Reverse-engineer

Suppose $n = 75$, σ is unknown. Construct a 95% confidence interval for μ.

Example 12: Confidence Interval for a Proportion

A sample of $n = 100$ one-gallon milk jugs is inspected. Of these, $X = 80$ contain at least one gallon of milk. Let p denote the population proportion of jugs that contain at least one gallon.

(a) Estimate the proportion of jugs that our filling process successfully fills with at least one gallon of milk.

(b) Construct a 95% confidence interval for p and make comments.

(c) Based on the confidence interval, is there sufficient evidence to conclude that the filling process successfully fills with at least one gallon of milk?

(d) If the sample had included 800 successfully filled jugs out of 1,000, the 95% confidence interval would have been _____.

Sample Size and Margin of Error

If we want the margin of error to equal a specified value B, what sample size is necessary?

$$B = z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}} \quad \Longrightarrow \quad \boxed{n = \hat{p}\hat{q}\left(\frac{z_{\alpha/2}}{B}\right)^2}, \qquad \hat{q} = 1 - \hat{p}.$$

If no prior estimate of p is available, use $\hat{p} = 0.50$.

Always round the required sample size **up** to the next whole number.

(d) Using the estimated proportion from part (a), what sample size is required for a 99% confidence interval for p with a margin of error of 0.01?

(e) If no prior estimate of p is available, what sample size is required for a 99% confidence interval for p with a margin of error of 0.01?